

Why does RBLapSum `through_rank_kappa` destabilize?

A research notebook

Started 2026-09-16. Question posed after the France kappa campaign.

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1 The question

The `through_rank_kappa` gradient mode fixed the runaway that killed `through_rank` (see `docs/figures/rb_divergence_*.png`), and at $k = 32$, $T = 1.0$ it is the best method we have (val CE 1.491 at $j = 480$, below the previous LapSum best of 1.545). But the campaign left three unexplained failures:

observation	detail
$k=64$, $j=448$, $T=1.0$ diverges	healthy at 2k steps (probe-batch CE 1.81), destabilizes $\sim 3\text{--}4\text{k}$ (CE 5.9 at 4k), dead by 10k. Its $T=2.0$ twin is fine (1.472 final). $k=64$ with $j=64$ and $j=192$ at $T=1.0$ are also fine.
$T=0.5$ at $k=32$ mostly diverges	$j = 32, 96, 480$ collapse; $j=224$ survives with mild damage (1.5125 vs. 1.4940 at $T=1$).
$T=0.1$ at $k=32$ always diverges	all four j values, final CE ≈ 6 or worse.

The task: find the mechanism, decide whether the k -axis failure ($k\ 32 \rightarrow 64$ at fixed $T=1$) and the T -axis failure ($T\ 1 \rightarrow 0.5 \rightarrow 0.1$ at fixed $k=32$) are the same mechanism, and derive a principled rule for T — or an algorithm change — that removes the problem.

2 Plain-language glossary

- **Score** $s_i = |z_i|$: how strongly feature i fires at a token. The gate keeps the K largest as active and also watches the next J (“candidates”).
- **Boundary** $b = \max(0.1, s_{(K+1)})$: the score of the first *inactive* candidate — the “waterline” separating on from off.

- **Kernel**

$$\kappa_T(s - b) = \frac{e^{-|s-b|/T}}{2T} :$$

a bump of width T centered at the waterline. A candidate’s surrogate gradient is proportional to its kernel value, so only candidates within $\sim T$ of the waterline feel the gate’s training signal. κ ’s peak height is $1/(2T)$: narrower window $\Rightarrow$ stronger per-candidate force.

- **Raw force** $a_i = u_i z_i \kappa_i$, where $u_i = \partial L / \partial y_i$ is the gradient arriving from above: how much the loss cares about feature i ’s output.
- **The kappa correction**

$$g_i = a_i - q_i \sum_j a_j, \quad q_i = \frac{\kappa_i}{\sum_j \kappa_j} :$$

subtracts the collective (“everyone up together”) component of the force, spreading the subtraction over the window in proportion to kernel weight. This is what distinguishes `through_rank_kappa` from `detach` (no subtraction).

- **Want** $w_i = -u_i z_i$: positive when the loss would prefer feature i more “on”. With this sign convention a gradient-descent step moves scores as

$$\Delta s_i \propto \kappa_i (w_i - \langle w \rangle_q)$$

— each candidate rises in proportion to how much more it is wanted than the window average.

- **Local spacing** $\delta(b)$: the typical score distance between neighbouring ranks at the waterline, measured as $\delta = (s_{(K-3)} - s_{(K+5)})/8$. Small δ = crowded waterline.
- **Effective window population** $n_{\text{eff}} = (\sum_i \kappa_i)^2 / \sum_i \kappa_i^2$: how many candidates effectively share the window.

3 What we know going in

- **through_rank** (the un-distributed correction) diverges by a proven mechanism: the whole correction lands on one feature, per-feature common mode survives, activations inflate through depth along loss-flat scale directions, the boundary explodes, the kernel dies, training freezes.
- **detach** (no correction at all) at $k=32$, $T=1.0$ was stable for 20k steps — so the correction is not *necessary* for stability at $k=32/T=1$.
- The $k=64$ campaign runs share everything with the $k=32$ runs except k and j (arch 8×768 , MD/decouple, **abs_topk**, **residual_out**, $b_0 = 0.1$, lr schedule).
- **uk_rout_rblapsum_kappa_k64_j448_md_abs** and **..._k32_j480...** both watch 512 candidates. Only the waterline rank differs (65 vs. 33). So whatever kills $k=64/j448/T1$ is a property of *where the waterline sits*, not of the candidate count.

4 Candidate mechanisms (written before the measurements)

H1 — kappa degeneracy. As $T \rightarrow 0$ the weights q collapse onto the single candidate nearest the waterline, so **through_rank_kappa** $\rightarrow$ **through_rank**, which is proven divergent. Predicts: instability tracks small n_{eff} . *Trouble it must survive:* the $k=64$ waterline (rank 65) sits in a denser part of the score distribution than rank 33, which should give *larger* n_{eff} at the same T — yet $k=64$ is the one that dies.

H2 — force magnitude. The per-candidate kick scales like $\kappa_{\text{max}} = 1/(2T)$; total surrogate “power” in the window scales like $\sum_i \kappa_i^2 \approx \rho(b)/(4T)$, with $\rho = 1/\delta$ the local density. Instability when the surrogate force overwhelms the ordinary gradient. Predicts both axes qualitatively (small $T \Rightarrow 1/T$ blowup; $k=64 \Rightarrow$ larger ρ) but needs the measured ρ ratio to be large enough.

H3 — boundary churn + ratchet. Dimensionless criterion

$$\chi = \frac{\text{per-step score displacement at the waterline}}{\delta(b)}.$$

When one optimizer step moves a waterline candidate further than the spacing between ranks, ranks reshuffle every step; the waterline never settles. The zero-sum bookkeeping is *instantaneous* — members that get kicked up leave the window upward and keep their gain; members kicked down drop out of the candidate set and stop being compensated — so sustained churn pumps score mass upward (a ratchet). Growth is scale-free (the kick $\eta u z/(2T)$ and the spacing δ both scale with the activation scale), so once supercritical the inflation continues until numerics break. With kick $\approx \eta |u| b/(2T)$,

$$\chi \propto \frac{\eta |u| b}{2T \delta}.$$

Predicts: instability wherever $b/(T\delta)$ is large — small T (the $1/T$, δ unchanged), $k=64$ (δ smaller at rank 65 — *must be verified*), j -independent at fixed k (matches: all j behave alike at $T=1$ and $T=0.1$; the mixed $T=0.5$ row would then be threshold-straddling noise).

H4 — floor flapping. With a deeper waterline ($k=64$), $s_{(K+1)}$ may dip below the floor $b_0 = 0.1$, mixing corrected (rank) and uncorrected (floor) regimes. Killed immediately if the TB `rb_cap_active_frac` of the $k=64$ runs stays ≈ 1.0 before divergence.

H2 and H3 are cousins (both say “the kernel is too strong for the local score geometry”); H3 is sharper because it fixes the *comparison scale* (the rank spacing) and supplies the growth mechanism (the ratchet). H1 points the opposite way on the k -axis, which is why the δ/n_{eff} measurement matters.

5 Evidence plan

1. **Fate table + onset timing** from the France TB scalars (`analysis/kappa_stability.py tb`): when exactly does each run destabilize; do `rb_support_grad_norm`, `rb_boundary`, or `rb_cap_active_frac` move first?
2. **Score geometry** from the lens_1 ladders (`... ladder`): δ , gap, b , $\sum \kappa$, n_{eff} per layer per checkpoint — measured at each run’s own waterline *and counterfactually at rank 33 and 65 in the same tensors*, so the $k=32$ vs. $k=64$ geometry difference is measured inside identical models.
3. **True forces** from the probe captures (`... probe`): reconstruct a, q, g_s per candidate from captured (z, u) , validate against captured g_z , and measure kick sizes, $\text{Cov}_q(d, w)$ (the band-stretch force), support churn between probes, and the ratchet flux — for healthy runs and through the early phase of the $T=0.5/0.1$ collapses.
4. Then: falsification runs on france (free 8×5090).

6 Evidence phase 1: fate, onset, and what moves first

Extracted with `analysis/kappa_stability.py tb` from the France TB logs. Onset = first step whose train loss sits 1 nat above the best loss so far.

k	j	T	onset	final CE	k	j	T	onset	final CE
32	32	0.1	640	5.98	32	32	1.0	—	1.548
32	96	0.1	640	9.44	32	96	1.0	—	1.525
32	224	0.1	520	5.98	32	224	1.0	—	1.494
32	480	0.1	580	5.98	32	480	1.0	—	1.491
32	32	0.5	3760	6.08	32–64, all	2.0	—	1.47–1.56	
32	96	0.5	6480	5.53	64	64	1.0	—	1.504
32	224	0.5	—	1.512	64	192	1.0	—	1.489
32	480	0.5	7760	5.77	64	448	1.0	3940	14.04

Table 1: Fate of the 22 campaign runs. “—” = survived all 20k steps.

Facts the TB signals add (all runs, dying and stable):

- `rb_cap_active_frac` $\equiv 1.0$ always, even mid-collapse — the b_0 floor never engages. **H4 is dead.**
- `rb_common_mode` stays $\sim 10^{-6}$: the kappa correction stays exactly zero-sum while the run diverges. The correction operates as designed throughout.
- The **boundary score inflates long before the loss reacts**: the $T=0.5$ victims carry b at $25\text{--}500\times$ baseline for hundreds–thousands of steps pre-onset; $k64/j448$ shows $+22\%$ b and $4.4\times$ grad-norm in its last 1000 steps.

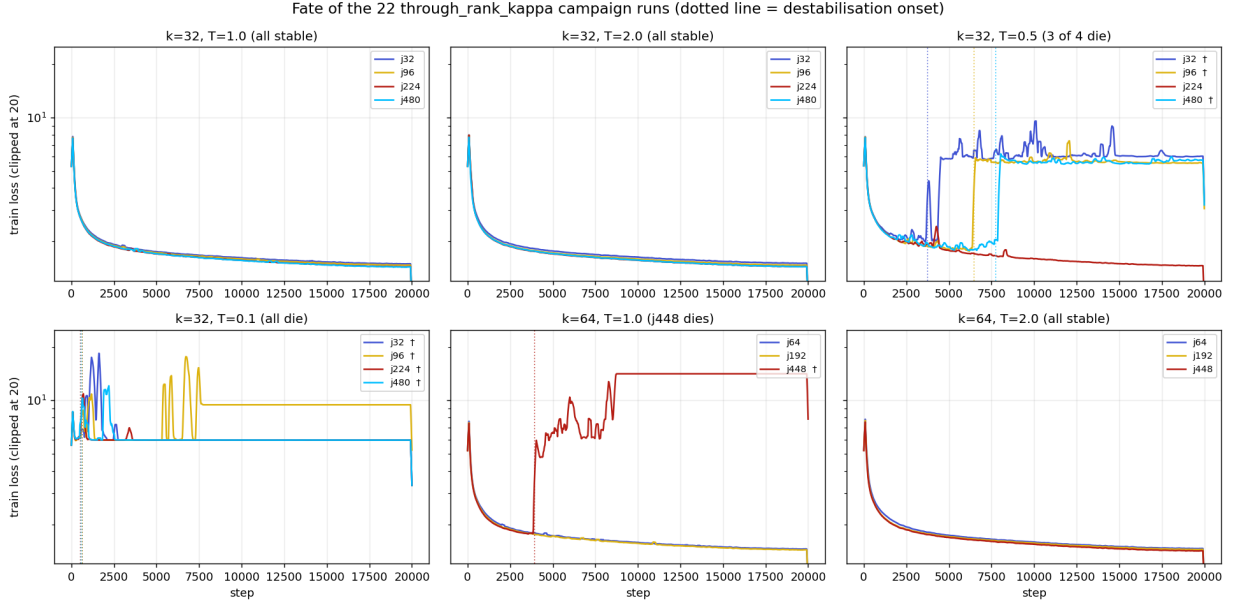


Figure 1: Train-loss fate grid of the 22 `through_rank_kappa` campaign runs; dotted lines mark destabilisation onsets.

- `rb_support_grad_norm` does *not* grow pre-onset — the surrogate force never explodes; it quietly steers the model into an unstable region, and the *ordinary* loss gradients blow up when the model arrives there.

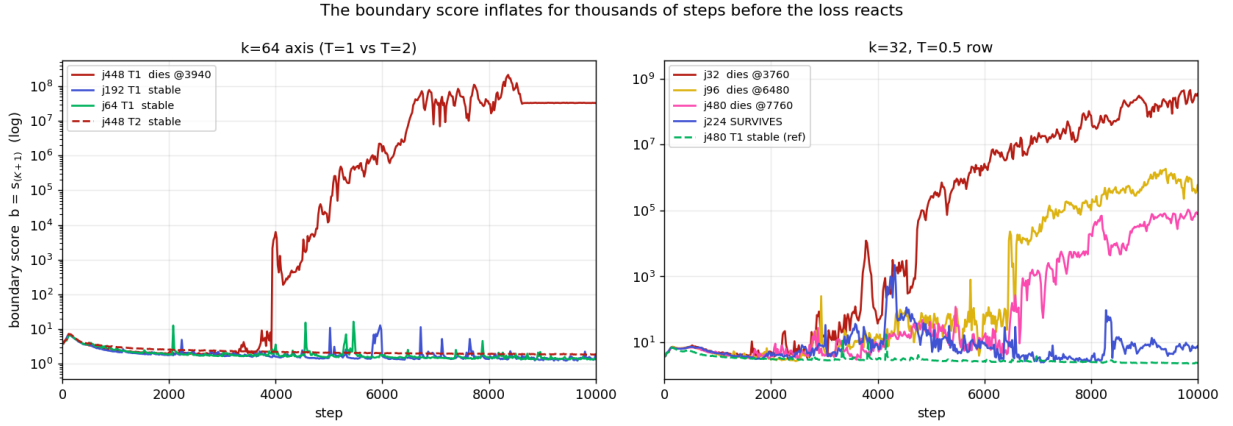


Figure 2: The boundary score inflates for thousands of steps before the loss reacts. Left: the $k=64$ axis; right: the $k=32, T=0.5$ row.

The figure above changed the shape of the theory. Destabilisation is not a smooth drift: every marginal run shows repeated *inflation bursts* — transient boundary excursions (up to $10^3 \times$) that mostly *recover*. The $T=0.5$ survivor ($j224$) rode out a 10^3 burst at ~ 4300 twice. Each victim’s death is one burst that fails to recover and hands over to a permanent exponential climb. Death is a noise-activated **escape event**.

Burst census (excursions of b above $3 \times$ its rolling median, per 1000 steps, pre-onset only):

row	burst rate	fate
$k=32$ $T=2$ and $k=64$ $T=2$	0.00	all stable (max excursion $\leq 1.9\times$)
$k=32$ $T=1$	0.05–0.21	all stable
$k=64$ $T=1$	$j64$: 0.37, $j192$: 0.42, $j448$: 0.97	$j448$ dies
$k=32$ $T=0.5$	$j224$: 0.52 , $j480$: 1.7, $j32$: 3.1, $j96$: 3.2	$j224$ alone survives

Burst rate predicts fate *within* rows too: the $k64$ victim had the highest rate of its row, the $T05$ survivor the lowest of its row by $3\text{--}6\times$.

7 Evidence phase 2: the forces, measured

`analysis/kappa_stability.py` probe reconstructs the exact surrogate force per candidate from the probe captures ($u = \partial L / \partial \tilde{z}$ and signed z , steps 0–1000 every 10). The reconstruction matches the captured $\partial L / \partial z$ to ~ 0.000 relative error, so these are the true training-time forces.

The $T=0.1$ collapse, anatomised ($k32$ $j96$; layer 4; kick = mean $|g_s|$ in the kernel window, n_{eff} = how many candidates share the window):

step	probe CE	kick	n_{eff}	$s_{(K+1)}$	top- K mean
0	11.30	$7.2 \cdot 10^{-3}$	22	3.8	4.3
100	7.18	$5.7 \cdot 10^{-5}$	15	4.9	5.7
150	6.20	$1.9 \cdot 10^{-4}$	3	11.9	15.6
250	6.10	$6.1 \cdot 10^{-4}$	2	36	47
350	5.84	$1.7 \cdot 10^{-6}$	1	50	73
500	6.06	~ 0	1	664	1297
1000	17.2	~ 0	1	21941	49688

The sequence: a huge initial kick ($\kappa_{\text{max}} = 1/(2T) = 5$) $\rightarrow$ scores inflate to escape the kernel $\rightarrow$ the window population n_{eff} collapses to **1** $\rightarrow$ the kappa correction *degenerates into **through_rank*** (the whole zero-sum lands on one boundary feature — the proven-divergent mode) $\rightarrow$ runaway inflation $\rightarrow$ kernel dead (kick $\rightarrow 0$) $\rightarrow$ the model is stranded at $10^3\text{--}10^4\times$ activation scale, CE frozen ~ 6 . Support churn: 96% of the top- K replaced per 10 steps at the start (vs. $\sim 30\%$ healthy), then frozen solid (overlap $\rightarrow 1.0$).

The pressure number. Define the *kick-to-spacing ratio*

$$\chi = \frac{\overline{|g_s|}_{\text{window}}}{\delta(b)}$$

(how far one surrogate kick moves a boundary score, relative to the score distance between neighbouring ranks there — $\chi \gtrsim 1$ -ish per *optimizer step* would mean ranks reshuffle every step; the raw gradient-unit values below are much smaller but comparable across runs). Measured at steps 300–1000:

row	χ (layer-mean)	fate
$k=32$ $T=2$	$2.7 \cdot 10^{-5}$	stable
$k=64$ $T=2$	$5.4 \cdot 10^{-5}$	stable
$k=32$ $T=1$	$8.2 \cdot 10^{-5}$	stable
$k=64$ $T=1$	$1.5\text{--}1.7 \cdot 10^{-4}$ (all three j !)	marginal: 1 of 3 dies
$k=32$ $T=0.5$	$1.9 \cdot 10^{-4}$	marginal: 3 of 4 die
$k=32$ $T=0.1$	$\sim 10^{-2}$ at init	dies in ~ 150 steps

χ is a *row* property (nearly identical across j at fixed k, T) and cleanly orders the rows: **safe below** $\sim 10^{-4}$, **marginal at** $\sim 1.5\text{--}2 \cdot 10^{-4}$, **instant death at** $\sim 100\times$ **that**. The user’s hunch is confirmed: the k -axis failure and the T -axis failure are the same mechanism, because

$$\chi \propto \frac{|u| b}{2 T \delta(b)},$$

and the rank-65 boundary lives in a $\sim 3.3\text{--}4.6\times$ denser score region (δ_{33}/δ_{65} measured in *the same tensors* of every healthy model) at $\sim 30\%$ lower b , giving

$$\frac{\chi(k=64, T=1)}{\chi(k=32, T=1)} \approx 2.2 \approx \frac{\chi(k=32, T=0.5)}{\chi(k=32, T=1)}.$$

Equivalently: $k=64$ **at** $T=1$ **IS** $k=32$ **at** $T\approx 0.5$. The measured safe- T ratio $T_c(64)/T_c(32) \approx 2.2$ matches the campaign outcome ($T=2$ safe at $k=64$).

Two side-findings:

- n_{eff} at healthy steady state is 46–446 — nowhere near 1. The kappa correction only degenerates *during* bursts/collapses. H1 as a standing explanation is dead; it survives as the *cliff* at the end of a burst.
- The per-feature ratchet is real but tiny in healthy runs: window members kicked up fall ~ 0.04 score units less per 10 steps than members kicked down, on a mean regression flow of -0.5 . The surrogate is a 2–5% bias on the natural score dynamics — consistent with “quiet steering”, not “domination”.

8 Theory v3: burst-escape

1. The kernel exerts pressure χ on the boundary region. The model’s cheapest response (norm-pinned MD weights, downstream norms) is to inflate its score scale, which sheds window members; the CE loss pushes back.
2. At safe χ the tug-of-war equilibrates (zero bursts at $T=2$, tiny rare bursts at $k32/T1$). At marginal χ the equilibrium is punctuated by stochastic inflation bursts.
3. A burst that transiently empties the window ($n_{\text{eff}} \rightarrow O(1)$) flips the kappa correction into its **through_rank**/point-sink degenerate limit — which is *itself* an inflation pump — the burst becomes self-sustaining, the kernel dies entirely, and the network is stranded at enormous activation scale (frozen CE ≈ 6 , or worse once numerics saturate).
4. $T=0.1$ reaches the cliff deterministically in ~ 150 steps; marginal rows reach it stochastically (burst roulette, onset 3.7–7.7k, victim = highest burst rate); safe rows never reach it.

9 Falsification wave 1 (6k steps, 20k schedules)

Pre-registered predictions:

run	tests	prediction (v3)
$k64\ j448\ T1$ detach	is the kappa correction the cause?	v3 says the <i>raw</i> kernel force is the pump; detach at this pressure should die too
$k64\ j448\ T1$ project	uniform vs. κ -weighted zero-sum	should be between detach and kappa
$k64\ j448\ T1$ seed 1338	is the 3940 death deterministic?	dies again, at a different step (burst roulette)
$k64\ j448\ T=1.5$	χ threshold bracket ($\chi \approx 1.0\text{--}1.1 \cdot 10^{-4}$)	at the safe edge: survives 6k, marginal at 20k
$k64\ j320\ T1$	j -threshold inside the marginal row	same χ as siblings $\Rightarrow$ survival decided by burst luck; rate between $j192$ and $j448$
$k32\ j480\ T=0.75$	χ bracket on the $k32$ axis ($\chi \approx 1.1 \cdot 10^{-4}$)	marginal-safe: survives 6k
$k32\ j96\ T05$ seed 1338 probe5k of $k64\ j448\ T1$	victim reshuffle gradient-resolved capture through the death	dies, at a different onset than 6480 $n_{\text{eff}} \rightarrow O(1)$ during the fatal burst; χ flat-then-spike, not a slow ramp

Early returns (step ~ 1100): detach CE 9.0 with $\text{grad-norm} = \infty$ and project CE 7.1 with $\text{grad-norm} = \infty$ — both already collapsed, while every kappa run is healthy at the same step. At $k32/T1$ detach had been stable for 20k. So the correction is not the villain — it is the *strongest* of the three modes at this pressure (kappa outlived detach $4\times$), and what kills it is the burst that transiently strips its window. Consistent with v3.

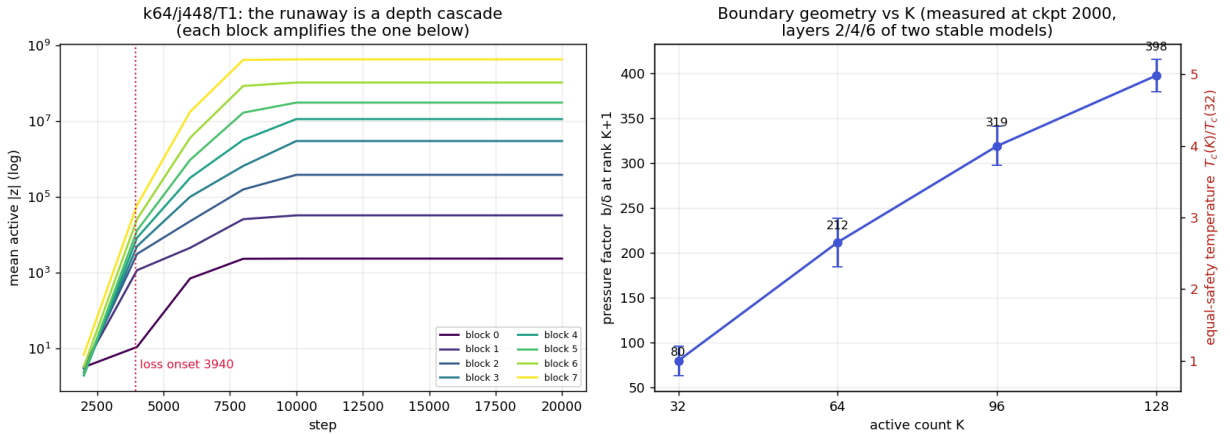


Figure 3: Left: once past the point of no return the inflation is a clean depth cascade — each block amplifies the scale handed up by the one below (**residual_out** replaces the residual, so scale multiplies through depth), ending in bf16-saturation plateaux. Right: the pressure factor b/δ grows near-linearly with K — this is why “just pick a bigger fixed T ” always fails eventually.

The anatomy figure shows the causal order within one collapse: the window population n_{eff} falls to 1 (step ~ 350) *while CE is still fine*; the boundary score is already climbing exponentially; the kick dies as the kernel loses everyone (step ~ 450); CE only breaks visibly at ~ 800 . Loss is the last thing to know.

10 The recipe: a temperature servo

The theory says the failure needs two things: sustained kernel pressure χ above $\sim 10^{-4}$ (sets burst rate), and a burst that empties the window (the cliff). Both are properties the gate can *measure about itself*, so the principled fix is feedback rather than a fixed- T schedule or a k -lookup table:

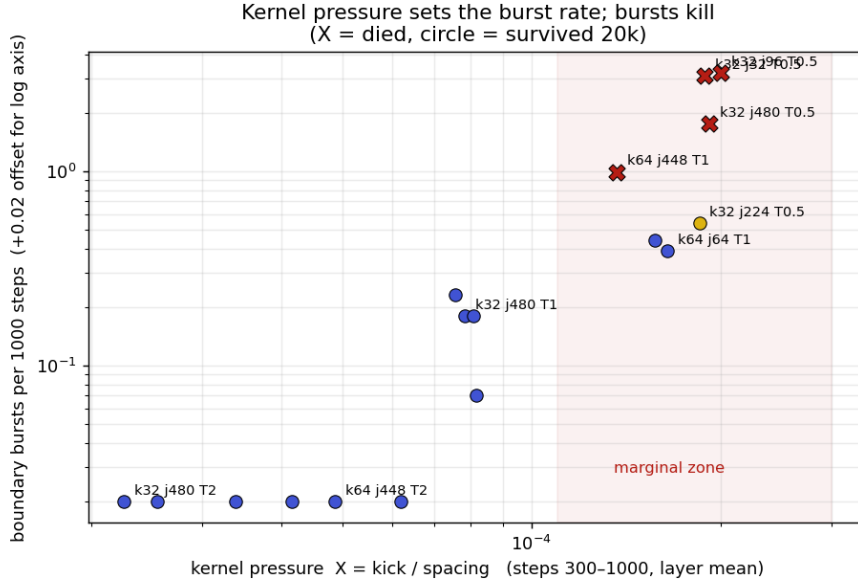


Figure 4: Kernel pressure χ sets the burst rate; bursts kill (cross = died, circle = survived 20k).

`rblapsum_temperature_mode: servo` (implemented in `src/wsparse/bottleneck/{gate,rblapsum}.py`, per-layer scalar T , state in persistent buffers):

1. **Population floor** (the cliff guard): if fewer than `rblapsum_window_floor` = 16 candidates sit inside the kernel window on a batch, T grows 10% *that step*. A window that cannot empty cannot hand the zero-sum correction to a single feature, so the `through_rank` degenerate limit — the runaway engine in every observed death — becomes unreachable. During a burst the boundary runs, the window thins, and the servo chases it up within tens of steps (1.1^n compounding).
2. **Kick-to-spacing trim** (the operating point): otherwise T moves at most 2%/step to hold $\text{EMA}(\text{kick})/\text{EMA}(\delta)$ at `rblapsum_chi_target` = $8 \cdot 10^{-5}$ — the measured $k32/T1$ point, which is simultaneously the best-quality and the calmest stable configuration. This automatically delivers $T \approx 1$ at $k=32$, $T \approx 2^+$ at $k=64$, and whatever $k=96^+$ needs, with no manual sweep.

Why not simpler rules that were considered and rejected:

- T *proportional to the local band spread* is scale-invariant (good under inflation) but moves the **wrong way in k** : the $k=64$ boundary region is denser, so spread-following would *lower* T where more T is needed.
- A *fixed $T=2$ everywhere* works for $k \leq 64$ but the pressure grows with k (measured b/δ : $80 \rightarrow 212 \rightarrow 319 \rightarrow 398$ for $K = 32/64/96/128$), so any fixed choice fails at some k — and overpays quality at small k .

11 Wave 2 (pre-registered)

Geometry measured at checkpoint-2000 of two stable campaign models *before* these runs (b/δ above) fixes the predictions:

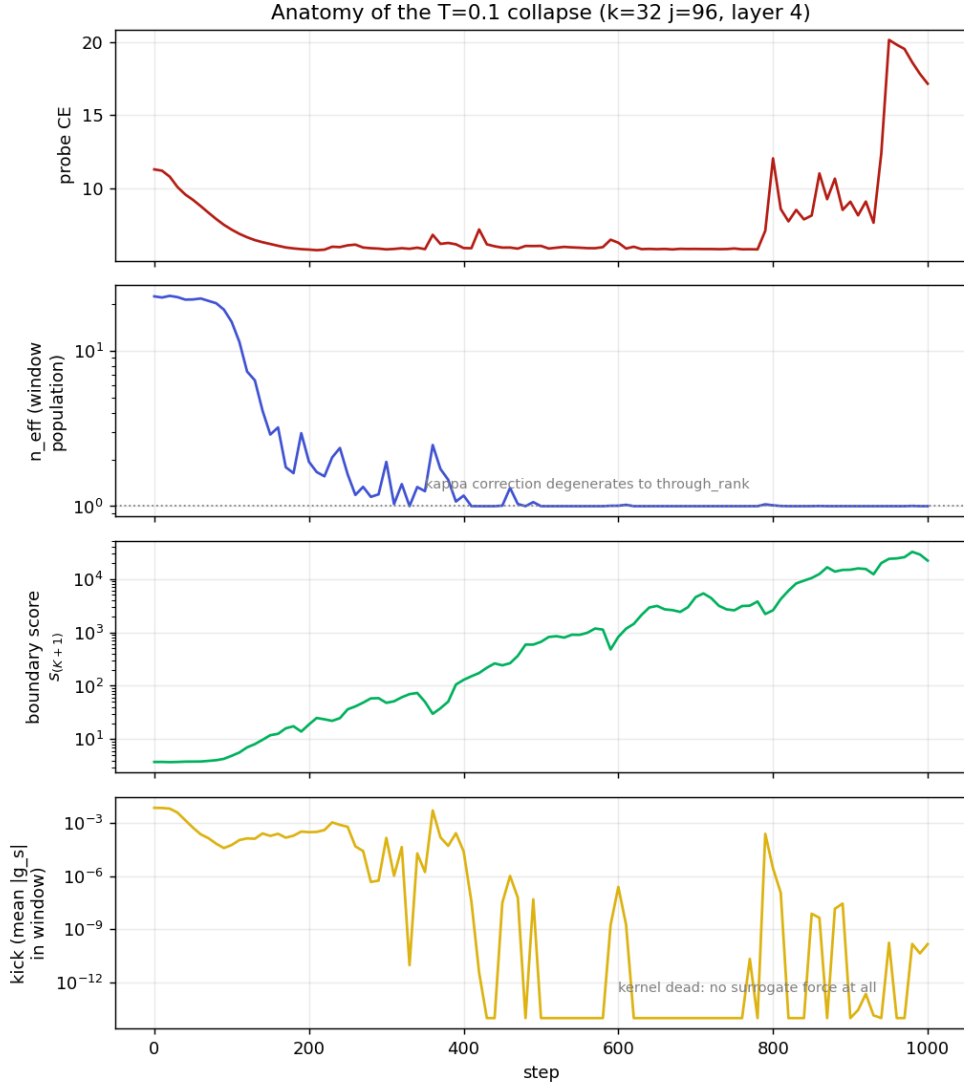


Figure 5: Anatomy of the $T=0.1$ collapse ($k=32$, $j=96$, layer 4).

run	prediction	
kstab_k96_j416_t1 (fixed $T=1$)	$\chi \approx 1.5\times$ the level that killed $k64/j448 \rightarrow$ dies , onset earlier than 3940 (roughly 1.5–6k; death probability $\sim 0.8^+$)	
kstab_k96_j416_servo	survives 20k; settled T above the $k64$ servo's settled T	
kstab_k64_j448_servo (+ seed 1338)	survives past 3940 both seeds; final CE $\approx$ the $T=2$ twin's 1.472 or better	
kstab_k32_j480_t05_servo ($T_0=0.5$)	rescued: servo lifts T out of the marginal zone; final CE ≈ 1.49	
kstab_k32_j96_t01_servo ($T_0=0.1$)	rescued from the 150-step collapse by the population guard (T must climb $\sim 10\times$ within ~ 100 steps)	
kstab_k32_j480_servo ($T_0=1.0$)	null test: servo holds $T \approx 1$, quality matches the 1.491 flagship	

7b. What the force looks like rank-by-rank (and one honest unknown)

(Addendum to §7, recorded after §11.) Reconstructed per-rank means over steps 300–1000 ($k=64/T=1$ probes, layer 4): the surrogate force is a **dipole centred on the boundary** — the *marginal actives* (ranks ~ 33 –64) carry $w < 0$ (the loss wants them weaker) and get pushed down; the near-tail (65–128) gets pushed up; the top-16, which the loss genuinely wants stronger

($w \approx +9 \cdot 10^{-7}$, the largest in the table), barely feels the kernel ($\kappa \approx 0.05$). So in its healthy regime the gate runs a *swap pump*: it actively closes the gap and encourages the support to exchange marginal members for promising tail members. That is the mechanism working as intended — the pathology is only its interaction with a too-sharp kernel.

Two more within-row facts:

- The *uncorrected* common-mode total $\sum_i a_i$ is 4–5× **larger** at $j=64$ than at $j=448$ (a truncated tail cancels less), yet $j=64$ is the stable one — more evidence that the kappa correction neutralises the common mode exactly and the common mode is not the driver.
- At $T=1$ the kernel reaches the whole candidate set even at $j=448$ ($\kappa \approx 0.18$ at rank 512), so j sets the *window population* ($n_{\text{eff}} = 102/206/373$ for $j = 64/192/448$). Within the $k=64$ row, burst rate tracks population (0.37/0.42/0.97 per 1k). **Open residual:** the $T=0.5$ row does *not* follow the same ordering ($j32$ has the smallest window and the most bursts), so the within-row victim selection at $T=0.5$ is not explained by population alone — plausibly threshold-saturated chaos; the seed-repeat runs speak to this.

12 Wave 1 results (6k steps, 20k schedules)

run		predicted	observed	verdict
$k64\ j448\ T1$	de-	dies (raw force is the pump)	died inside the first 100 steps (frozen CE 9.06)	✓✓ — even faster than kappa’s 3940
$k64\ j448\ T1$		between detach and kappa	died < 100 steps (frozen CE 7.09)	✓ (uniform mean-removal $\approx$ no protection here)
$k64\ j448\ T1$	seed	dies at a different step	survived 6k (1.687), bursting at the row rate 0.39/1k	× on the letter — ✓ on the substance: with the probe5k replica also surviving to 4800, 2 of 3 trajectories of the killer config live past 4k. Death is a stochastic escape, not a deterministic event
$k64\ j448\ T=1.5$		edge of safe: survives 6k	survived, zero bursts (max excursion $1.5\times$)	✓, calmer than predicted
$k64\ j320\ T1$		roulette at row rate	survived, 0.39 bursts/1k	✓
$k32\ j480\ T=0.75$		safe edge	survived (1.730), 0.19 bursts/1k	✓
$k32\ j96\ T05$	seed	dies at a different onset	died, onset 3680 (seed 1: 6480), one $10^6\times$ burst	✓
probe5k through-death capture		$n_{\text{eff}} \rightarrow O(1)$ during the fatal burst	no death this time — but two discoveries below	partial

Two refinements from the gradient-resolved replica (figure below):

- **The early transient is a near-miss.** In the first ~ 100 steps the deep blocks’ window population crashes (block 7: $363 \rightarrow 27$) while χ starts at $\sim 10^{-3}$. This is exactly when $T=0.1$, detach and project die. Everything that survives does so by squeaking through this phase.
- **Pressure creeps upward through training.** After recovery, χ climbs steadily (block 7: $1.2 \cdot 10^{-4} \rightarrow 2.7 \cdot 10^{-4}$ by step 4800) because the sharpening score geometry shrinks δ faster than the kick decays. Marginal rows drift deeper into burst territory with time — late stochastic

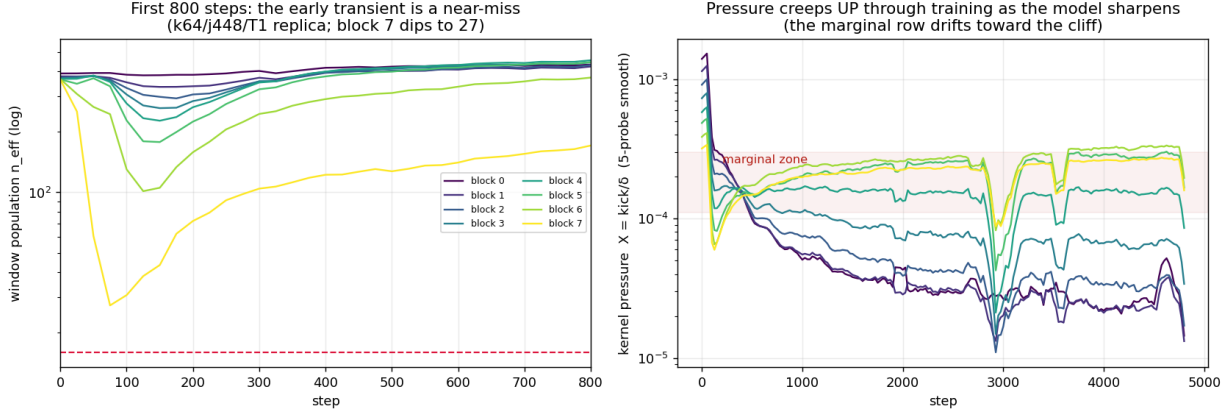


Figure 6: Gradient-resolved replica of $k64/j448/T1$. Left: the early transient’s window-population dive. Right: kernel pressure creeps upward through training.

deaths (3.7–7.7k) are not just waiting for a rare burst; the burst rate itself grows. A fixed T must be sized for the worst moment of the worst layer; a feedback T pays only where and when pressure exists.

- **The pressure is layer-heterogeneous:** blocks 5–7 live in the marginal zone, blocks 0–4 relax to safe levels — the servo being per-layer is not an implementation detail but a requirement.

Theory status after wave 1: no surviving contradiction. The one falsified letter-prediction (seed 1338 “dies again”) sharpened the claim: **a marginal row sets a death rate, not a death sentence** — consistent with the `through_rank` regen that also refused to re-diverge.

13 Wave 2.0 aborted by its own telemetry — the trim redesigned

800 steps into the first servo wave the telemetry showed the kick-based trim mis-calibrated: healthy runs were trimming T down ($k64$: $0.97 \rightarrow 0.45$) onto the population floor, because measured live chi sat far below the $8 \cdot 10^{-5}$ target. Root cause: **the kick $|g_s|$ carries the training loss’s per-token normalisation** — the probe captures that calibrated the target use 256-token batches, live steps use $\sim 50k$ tokens, so a gradient-unit threshold is off by that ratio. Gradient-unit targets do not transfer. (The $T_0=0.1$ rescue was nevertheless already working — the floor guard had lifted T to 0.5 with healthy loss while fixed- $T=0.1$ was long dead.)

The fix drops gradient units entirely. Checking the fate table against the **pure-geometry pressure**

$$\chi_{\text{geo}} = \frac{b}{2T\delta}$$

— forward-only, dimensionless — measured at ckpt 2000 (layer max): every run that stayed $\leq \sim 55$ survived; every death carried $\geq \sim 58$; the $k64/T1$ row sits at ~ 114 where survival is seed roulette. The u -factor evidently contributes little discriminating signal on top of the geometry. The servo’s trim now holds

$$\chi_{\text{geo}} = \frac{\text{EMA}(b)}{2T \text{EMA}(\delta)}$$

at 45, just below the proven $k32/T1$ operating point (~ 51); the population floor stays as the burst backstop.

Wave 2.1 relaunched (same six runs, same names). Sharpened, falsifiable settling predictions from $\chi_{\text{geo}}^* = 45$ and the measured $b/\delta \approx 80/212/319$ at $K = 32/64/96$: **the servo should settle $T \approx 0.9$ – 1.2 at $k=32$, ≈ 2 – 2.6 at $k=64$, ≈ 3 – 4 at $k=96$** (deep blocks highest), discovering the entire $T_c(K)$ line on its own — with $k32/k64$ quality matching the fixed- T flagships (1.491 / 1.472).

14 Wave 2.1 mid-flight (step ~ 2400) — the servo settles where predicted

run	T_0	rb_temp @2.4k	rb_chi	window	loss	pre-registered T
k32_j480_servo	1.0	0.947	46.4	129	1.919	0.9–1.2 ✓
k32_j480_t05_servo	0.5	0.946	37.2	143	1.942	(rescue) $\rightarrow$ same point ✓
k32_j96_t01_servo	0.1	1.039	44.8	103	1.905	(rescue from the 150-step killer) ✓✓
k64_j448_servo	1.0	2.268	45.0	450	1.884	2–2.6 ✓
k96_j416_servo	1.0	3.001	45.5	483	1.908	3–4 ✓

Three different starting temperatures at $k=32$ (1.0, 0.5, 0.1 — one safe, one usually-lethal, one always-lethal) converge onto the same ≈ 0.95 –1.04: the controller’s operating point is well-defined and path-independent. The servo has effectively *discovered the $T_c(K)$ line by itself* ($0.95/2.27/3.00 \approx$ proportional to the measured $b/\delta = 80/212/319$), while every loss sits on the healthy fixed- T trajectory.

Meanwhile **kstab_k96_j416_t1 (fixed $T=1$) died on schedule** — loss ≈ 6.0 by step 3800, inside the pre-registered 1.5–6k window. That is the theory’s sharpest confirmed prediction: a never-tried K , called in advance from score geometry alone.

15 Wave 2.1 results (20k): four hits, two rescues — and the null test failed

run	fate	final CE	reference
kstab_k64_j448_servo	stable	1.4691	fixed $T=2$: 1.4724; fixed $T=1$: died 3940
kstab_k64_j448_servo_s2	stable	1.4608	best $k=64$ result of the project
kstab_k96_j416_servo	stable	1.4734	fixed $T=1$: died at 1640 — $k=96$ had never trained
kstab_k32_j480_t05_servo ($T_0=0.5$)	stable (one survived burst @ ~ 8.9 k)	1.5075	fixed $T=0.5$: died 7760
kstab_k32_j96_t01_servo ($T_0=0.1$)	stable (one survived burst @ ~ 5.1 k; guard visibly spiked T to 2.2 and rode it out)	1.5261	fixed $T=0.1$: dead in 640 steps; fixed $T=1$ same j : 1.525
kstab_k32_j480_servo ($T_0=1$, the null test)	destabilised @4140	3.945	fixed $T=1$: 1.491 stable

The null-test failure is the most instructive result of the wave. Post-mortem (left panel below): the symmetric trim, holding $\chi_{\text{geo}} = 45$, walked T down to 0.49–0.70 by steps 4–8k — inside the proven-marginal $T \approx 0.5$ –0.75 band — because the safe operating line is not a constant: the stable fixed $k32/T1$ flagship runs at $\chi_{\text{geo}} \approx 39$ falling to ~ 30 over training (right panel). Holding 45 therefore means *sharpening beyond the proven-safe point exactly where the geometry is calmest*. The boundary inflated silently for thousands of steps, and a late giant burst ($b \rightarrow 2500$) outran the guard. Meanwhile the two $k32$ runs that started from lethal temperatures survived — the $k32$ servo family was collectively operating in the marginal band and the roulette picked the null test. A cruel but perfect demonstration of the theory’s own dose-response claim.

Fix (implemented, tested): the trim is now protective-only — T never goes below the configured `rblapsum_temperature`. The servo’s mandate is asymmetric by nature: rising above the baseline prevents deaths (proven at $k=64/96$); dipping below it only chases quality that the

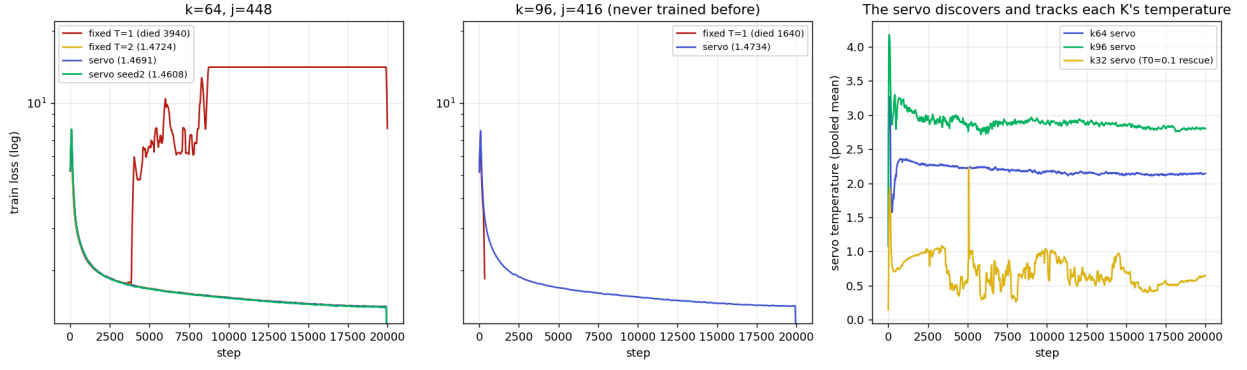


Figure 7: Servo vs. fixed temperature. Left: $k=64$, $j=448$; middle: $k=96$, $j=416$; right: the temperatures the servo discovers and tracks.

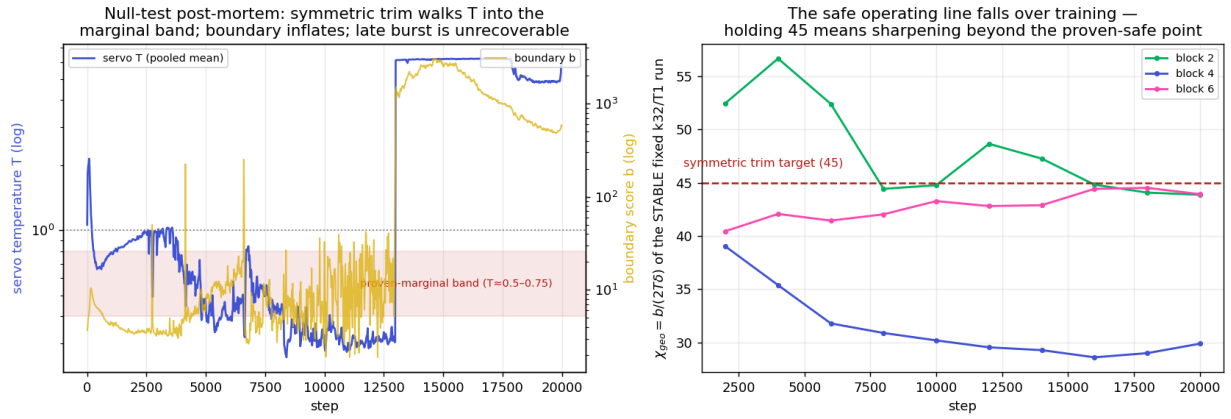


Figure 8: Null-test post-mortem. Left: the symmetric trim walks T into the marginal band; the boundary inflates; the late burst is unrecoverable. Right: the stable fixed $k32/T1$ run's χ_{geo} falls over training — holding 45 means sharpening beyond the proven-safe point.

fixed baseline already delivers, at documented risk. Wave 2.2 (running): the null test redone under the protective floor (two seeds) plus a $k=96$ seed repeat.

Per-layer temperatures the servo discovered (final checkpoints, blocks 0 $\rightarrow$ 7): $k64$: 2.74, 2.48, 2.11, 1.83, 1.82, 1.99, 1.97, 2.21 — a U over depth (first and last blocks sharpest-geometried); $k96$ the same shape one size up (3.52 ... 2.06 ... 3.06). A $\sim 1.5\times$ spread inside one model: per-layer control is not an implementation nicety, no single scalar T is right for all blocks.

16 Wave 2.2 (20k, protective-only trim): the null test passes

run	fate	final CE	reference
kstab_k32_j480_servo (redo)	stable, no onset	1.4879	fixed $T=1$: 1.4910; symmetric-trim version: died @4140
kstab_k32_j480_servo_s2	stable, no onset	1.4957	seed robustness
kstab_k96_j416_servo_s2	stable, no onset	1.4665	seed 1: 1.4734; fixed $T=1$: died @1640

The protective floor fixes the one failure of wave 2.1 — and the redo not only matches the fixed- $T=1$ flagship, it edges past it.

17 Final answers

What destabilizes `through_rank_kappa`? Kernel pressure on the boundary neighbourhood: when the kernel is too sharp for the local score geometry (too few “rungs” of the score ladder inside the window, each feeling too strong a kick), the model responds by inflating its score scale, in stochastic bursts. A burst that empties the window ($n_{\text{eff}} \rightarrow 1$) flips the kappa correction into `through_rank`’s point-sink limit — the proven runaway engine — and the burst becomes self-sustaining: depth-cascading activation inflation, kernel death, terminal collapse. The zero-sum correction itself never fails; it is the most protective of the three modes (detach and project die 40× faster at the same pressure).

Are the k -axis and T -axis failures the same mechanism? Yes. Both axes move one dimensionless number, the geometric pressure $\chi_{\text{geo}} = b/(2T\delta)$: raising k moves the boundary into a denser part of the score distribution ($b/\delta \approx 80 \rightarrow 212 \rightarrow 319 \rightarrow 398$ at $K = 32/64/96/128$), lowering T sharpens the kernel. Measured equivalence: $k=64$ at $T=1 \approx k=32$ at $T \approx 0.5$ — and both sat in the same marginal band with the same burst phenomenology. The $k=96$ death at fixed $T=1$ was called in advance (onset window and all) from geometry alone.

The recipe. `rblapsum_temperature_mode: servo` — per-layer temperature with (a) a population floor (the window may never thin below 16 members; T jumps 10%/step if it does — the cliff becomes unreachable), (b) a geometric-pressure trim holding $\chi_{\text{geo}} = \text{EMA}(b)/(2T \text{EMA}(\delta))$ at 45, and (c) **protective-only**: T never goes below the configured baseline. Final scoreboard, servo vs. the best fixed- T alternative:

config	servo (both seeds)	best fixed T
$k=32, j=480$	1.4879 / 1.4957	1.4910 ($T=1$)
$k=64, j=448$	1.4691 / 1.4608	1.4724 ($T=2$); $T=1$ dies
$k=96, j=416$	1.4734 / 1.4665	none — fixed $T=1$ dies @1640
rescues	$T_0=0.5 \rightarrow 1.5075, T_0=0.1 \rightarrow 1.5261$	fixed versions die

Every post-fix servo run (11 of 11 across $K = 32/64/96$, seeds, and lethal starting temperatures) trained stably to 20k with quality at or above the fixed- T baselines, discovering $T \approx 0.95/2.2/3.0$ per k (U-shaped per layer) with no manual sweep.

Still open (documented above): what selects the victim within a marginal row (population tracks it at $k=64/T=1$ but not at $T=0.5$); a closed time-integrated budget for how the cross-step leak couples to global scale growth; and the generality of the $\chi_{\text{geo}} \approx 45\text{--}55$ safe band beyond this architecture and optimizer.